NAME:……………………………………………………………….STREAM:………

ST. ANDREA KAAHWA'S COLLEGE-HOIMA

S.5 END OF YEAR EXAMINATION 2025

UGANDA ADVANCED CERTIFICATE OF EDUCATION

CHEMISTRY

2 HOURS :30 MINUTES

INSTRUCTIONS TO CANDIDATES

- Attempt **all items in section A** including any **three from section B**
- Write the responses in section A in the spaces provided.
- Begin each item in section B on a fresh page and number each item attempted.
- No extra items will be marked.
- Where necessary, use (H=1, C=12, O=16, Na=23)

SECTION A

Item 1

A Russian scientist reported that the mass of a radioactive isotope, **T** under investigation reduced by 32% in 40 days. Predict the half-life of **T** and use it to outline the impact of its presence in the environment and how it can be mitigated.

……
……
……
……
……
……
……
……
……
……

Item 2

Elemental analysis shows that an organic compound, **J** having a density of $2.05 \times 10^{-3} gcm^{-3}$ at s.t.p contains 52.2% carbon, 13.1% Hydrogen and the rest being oxygen. An isomer of **J** when dehydrated with concentrated sulphuric acid yields compound **K** which then gives a positive test with bromine water.

(1 mole of a gas at s.t.p occupies $22.4 dm^3$ **)**

a) Determine the molecular formula and the possible isomers of J.

....
....
....
....
....
....
....
....
....
....
....
....
....

b) Write an equation leading to the formation of **K** from one of the isomers of **J** and hence outline the suitable mechanism.

....
....
....
....
....
....
....

c) Highlight on the statement "**K** gives a positive test with bromine water"

....
....

Item 3

The table below shows the values of enthalpies of combustion of carbon, rhombic sulphur and liquid carbon disulphide.

Substance	**Enthalpy of combustion ($KJmol^{-1}$)**
Carbon	-393.3
Sulphur	-293.7
Carbon disulphide	-1108

Deduce the enthalpy of formation of carbon disulphide and comment on its stability.

....
....
....
....
....
....
....
....
....

Item 4

Explain the following laboratory observations:

a) When ammonia solution was added to magnesium sulphate solution, a white precipitate insoluble in excess was formed.

....
....
....
....
....
....
....
....

b) Beryllium chloride has a lower melting point than lithium chloride .

………
………
………
………
………
………
………

Item 5

In culture solutions, acetate buffer is used to maintain a suitable optimal P^H for microbal growth. Describe how you would prepare a buffer solution of P^H 5 using ethanoic acid. (Ka for ethanoic acid is $1.8 \times 10^{-5} moldm^{-3}$)

………
………
………
………
………
………
………
………
………
………
………
………
………
………
………
………

SECTION B

Item 6

In Jinja city, a new **metal-fabrication** workshop has recently opened. The workshop uses several metals which are **Period 3 elements** to produce electrical cables, metal frames, and fertilizer additives.

After a few months of operation, residents began to report several problems:

- ✓ Some storage containers exploded due to **metals reacting violently** when they accidentally came into contact with **water**.
- ✓ Workers developed **skin burns** because poorly disposed metal scraps reacted with **dilute sulphuric acid** that leaked from cleaning tanks.
- ✓ The drainage system got blocked by lumps formed when some of the metals reacted with **sodium hydroxide solution**.
- ✓ A section of the workshop caught fire when **some metals** reacted with **oxygen**.

The local authorities requested a chemistry student (you) to help the company management understand how different **metals which are being used for fabrication react with the above mentioned substances**, so that they may improve safety and waste disposal.

Task: As a chemistry student; make a scientific write up you would present to the local authorities and company management in an organised one day workshop with the possible mitigation

Item 7

An **Agro-Fertilizer Plant** produces **ammonia** used in making nitrogen fertilizers such as ammonium nitrate and urea. Ammonia is manufactured by the **Haber process**, where nitrogen from air reacts with hydrogen obtained from natural gas.

Recently, farmers across the region complained of **declining crop yields**, saying fertilizer supplies from the plant had become **scarce** and **expensive**. When the Ministry of Agriculture inspected the plant, engineers found several societal problems:

- ✓ The factory's **compressor unit was malfunctioning**, meaning the required high pressure could not be maintained.
- ✓ The **temperature control system was damaged**, often causing temperatures to rise far above the recommended level.
- ✓ Workers reported that the **iron catalyst chambers were becoming inefficient** due to impurities entering the system.
- ✓ Amount of **nitrogen produced had reduced** since the air separation unit had become faulty, so a small amount of it was being used to produce ammonia.

  These technical problems slowed ammonia production, increasing costs and causing shortages in local markets. The shortage forced farmers to use less fertilizer, contributing to low maize, bean, and rice yields, affecting food security.

The plant management in an aim to restore normal operation seeks help to understand how;

a) The above **equilibrium conditions of the Haber process** influenced the low ammonia production.
b) They can start manufacturing **ammonium sulphate fertilizers** using Sulphur dioxide as the starting material.

Task: Make a scientific write up you would present to the plant management. (where necessary, support your responses using chemical equations)

Item 8

In a rural Ugandan community, a local processing cooperative wants to manufacture small quantities of useful organic compounds for educational and industrial purposes. They have limited chemicals and need to ensure safety and efficiency. The cooperative has tasked you to synthesize several compounds from available starting materials using the chemical reagents and stating suitable conditions at each step

a) Benzene from ethanol
b) 1-bromopropan-2-ol from calcium dicarbide
c) Chloroethane to ethyne
d) 1,2-dichlorobutane to 1-chlorobutane
e) Ethene to diethylether

Item 9

In Tororo District, a ceramics and brick-firing factory uses various ionic compounds, especially barium chloride in glaze formulation and in hardening clay products. Recently, the factory began experiencing several serious problems:

- ✓ Some ceramic products were cracking during firing, causing financial losses.
- ✓ Workers complained that the new batches of barium chloride supplied from a cheaper distributor dissolved too slowly in water, affecting production time.
- ✓ Engineers suspected that the quality problems were due to differences in lattice energy between the genuine and counterfeit barium chloride being delivered.
- ✓ Community members also raised concerns when traces of barium compounds were found in wastewater due to improper disposal.

The factory management is in need of knowledge about **lattice energy**, **factors affecting it**, and how to determine the **lattice energy and enthalpy of solution of Barium chloride, $BaCl_2$**, so they can verify the purity and quality of the materials they use.

Thermochemical changes	**Energy Value ($kJ\ mol^{-1}$)**
Sublimation of Barium	+180
1st Ionization energy of barium	+503
2nd Ionization energy of barium	+965
Bond dissociation of Chlorine molecule	+244
Electron affinity of Chlorine	−349
Enthalpy of formation of Barium chloride	−858

The enthalpy for hydration of barium ions and chloride ion per mole are −1,360 $kJ\ mol^{-1}$ and −364 $kJ\ mol^{-1}$ respectively.

Task: Make a scientific write up giving a detailed response to the problems identified and the possible mitigation.

END